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A prospective study of risk factors for hamstring injury in Australian football league players

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ABSTRACT

This study aimed to explore the association between hamstring strength, age and lower limb soft tissue injury history and subsequent hamstring injury among Australian Football League (AFL) players. This prospective cohort study recruited 125 players from three professional AFL teams. Eccentric knee flexor strength was assessed while performing the Nordic hamstring exercise in pre-season, and injury data were collected retrospectively (hamstring, groin, calf, quadriceps and knee), and prospectively (hamstring injuries) for one AFL playing season. Fourteen players (11%) sustained a hamstring injury in the subsequent playing season. Nordic strength was not significantly associated with future hamstring injury (Odds Ratio (OR) 1.9, $p = 0.36$), whereas player age greater than 25 years (OR = 2.9, $p < 0.05$), report of a hamstring injury within the previous year (OR = 3.7, $p = 0.01$), or greater than 1-year (OR = 3.6, $p = 0.01$), a previous groin (OR = 8.6, $p < 0.01$) or calf injury (OR = 4.6, $p = 0.01$) were factors significantly associated with subsequent hamstring injury. Based on these findings, increasing age and previous hamstring, groin and calf injury are all associated with an elevated risk of subsequent hamstring injury in AFL players.

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KEYWORDS
Lower limb injury; nordic strength; calf injury; groin injury

Introduction

Hamstring injury has been the most common cause of lost playing time among professional Australian Football League (AFL) players for the past 20 years (Orchard et al., 2013). These injuries affect approximately one in eight players each season, result in approximately 20 missed matches per club (Orchard et al., 2013) and cost individual clubs upwards of 0.25 million dollars in wages paid to players who were unable to take the field (Hickey et al., 2014). In similar sports such as Gaelic football (Roe et al., 2018) and soccer (European football) (Ekstrand et al., 2016), the incidence of first-time hamstring injuries has increased over 8- and 13-year periods, respectively. An improved understanding of predisposing risk factors for hamstring injury is needed to inform the design of interventions aimed at reducing the burden of hamstring injury in sports associated with higher risk of these injuries (Finch, 2006; Van Mechelen et al., 1992).

Hamstring strength has been proposed as a potential modifiable risk factor for hamstring injury (Opar et al., 2015). Isokinetic dynamometry is considered the gold standard assessment of hamstring strength, although prospective studies examining the association between isokinetically derived concentric or eccentric hamstring strength and hamstring injury in the AFL have reported mixed results (Bennell et al., 1998; Cameron et al., 2003; Orchard et al., 1997), whilst a recent meta-analysis demonstrated no association between any isokinetic strength variables and hamstring injury risk (Green et al., 2020). An alternative field-based measure of hamstring strength, the Nordbord, was used to assess strength during the performance of the Nordic hamstring exercise (NHE) during the 2013 AFL playing season, Opar et al. (2015) demonstrated players with Nordic strength less than 279 N were 4.3 times more likely to sustain a future hamstring injury than their stronger counterparts. However, in a subsequent study conducted over the 2013 and 2015 AFL seasons, Ruddy et al. (2018b) demonstrated that Nordic strength, age and previous hamstring injury could not be used to identify athletes at increased risk of hamstring injury with any consistency. Ruddy et al. (2018b) further reported that Nordic strength had increased on average by 31% in the 2015 season compared with the 2013 season. These strength differences may reflect changes in the training practices of AFL players over this time, particularly an increased emphasis on hamstring conditioning, and could explain the reported lack of association with hamstring injury. Given the reported increase in Nordic strength in the AFL observed in recent years (Opar et al., 2015; Ruddy et al., 2018b) and evidence for an apparent trend towards further increases in hamstring strength in the AFL (Vald Performance, 2019), additional investigation of the relationship between Nordic strength and hamstring injury in the AFL appears warranted.

Non-modifiable factors such as player age and injury history may also influence a player's risk of future hamstring injury (Gabbe et al., 2006; Orchard, 2001; Orchard et al., 2020; Verrall et al., 2001). AFL players older than 23 years were reported to be 1.3 to 4.4 times more likely to sustain a future hamstring injury than younger players (Gabbe et al., 2006; Orchard, 2001; Verrall et al., 2001). However, a more

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recent study did not report an association between increasing age and future hamstring injury, and the authors proposed that the explanation for this finding was due to the interaction between age and previous hamstring injury (Orchard et al., 2020). In contrast, there is consistent evidence that prior hamstring and other lower limb soft tissue injuries are associated with future hamstring injury in AFL players (Orchard, 2001; Orchard et al., 2020; Verrall et al., 2001). In the largest of these studies, which reported 1932 hamstring injuries across 272,759 player matches, AFL players with a recent history of hamstring injury were up to 13 times more likely to suffer a recurrence than those without a history, with a greater risk apparent within the first 15 weeks of return to play (Orchard et al., 2020). Furthermore, other prior injuries to the lower limb, such as calf (Orchard, 2001; Orchard et al., 2020) and pubic-related groin (Verrall et al., 2001) injuries were also associated with increased hamstring injury risk, although a lack of association has also been reported (Opar et al., 2015). It has been suggested that other lower limb injuries may alter biomechanics, thereby increasing the risk of future hamstring injury (Orchard, 2001; Verrall et al., 2001); however, more research is needed to clarify this. As the abovementioned factors have the potential to change with training practice and corresponding hamstring strength, further investigation of these risk factors is also warranted.

The incidence of hamstring injury in the AFL remains high (AFL, 2018; Orchard et al., 2013) and there is uncertainty about the role of hamstring strength in hamstring injury prevention and how this might be affected by changing training practices. The purpose of this study was therefore to prospectively assess whether pre-season measures of Nordic strength, player age and a history of hamstring, and other lower limb soft-tissue injuries, were associated with subsequent hamstring injury in an independent sample of AFL players. We hypothesized that (1) older age and injury history would be associated with subsequent hamstring injury; and (2) based on recent literature, Nordic strength would not be associated with subsequent hamstring injury.

Methods

Participants

Professional male AFL players from three clubs were invited to participate in this study. Of the 142 eligible players, 17 did not complete testing due to unavailability at the time of testing, which resulted in 125 players being included. The age, height and mass of the players (mean, standard deviation; SD) were 22 (3) years, 189 (7) cm and 87 (8) kg, respectively. Sample size was calculated a priori with input parameters (effect size = 0.8, alpha = 0.05, power = 0.8) used for a two-tailed independent samples *t*-test (G*Power version 3.1.9.4). An allocation ratio of 8 was used for the ratio of two groups, which is based on the incidence of hamstring injury within the AFL (Orchard et al., 2013). This yielded a total sample size of 128 subjects, with group sizes of 114 and 14. The study was approved by the Human Research Ethics Committee (2017/896) of the relevant institution and all participants provided written informed consent.

Baseline measures

All participants completed a standardized questionnaire at the mid-to-end of an AFL pre-season (January to March, n = 98 in 2017 and n = 27 in 2018). This was administered by a sports physiotherapist, and assessed their history of hamstring injury (less than or greater than 1-year ago), and any history of groin, quadriceps, knee (including anterior cruciate ligament) and calf injuries that resulted in at least one AFL match missed. Bilateral Nordic strength was assessed by trained club personnel using a Nordbord (Vald Performance, Brisbane, Australia), using a standardized protocol (Opar et al., 2013, 2015). Shank length was measured from the lateral knee joint line to the distal aspect of the fibula. Prior to the performance of the NHE test, participants performed a warm-up consisting of less than five submaximal repetitions of the NHE, followed by three maximal repetitions. All players regularly performed the NHE as part of hamstring injury reduction programs, and hence were familiar with the NHE prior to testing. The instructions to the participants were to maintain neutral pelvic and hip alignment and to slowly lean forward while resisting the forward lean by maximally contracting the hamstrings. All trials were monitored by club personnel to ensure adherence to proper technique and trials were repeated if correct technique was not adhered to. A trial was deemed acceptable when the force output reached maximum, which was followed by a sharp decline in force which occurred when the player was no longer able to resist the effects of gravity acting against the segment above the knee joint. A between session reliability study of hamstring strength assessed using a prototype of the Nordbord used in the present study has been reported to be high (Intraclass correlation coefficients of 0.83 to 0.90, typical error = 21.7 to 27.5 N (5.8–8.5%)) (Opar et al., 2013). Force data from both limbs were sampled at 100 Hz and transferred to a cloud-based software program (ScoreBord, Vald Performance). Nordic strength was determined as the maximum force generated from the three maximal repetitions and was reported in absolute terms (Newtons, N). Data were coded according to dominant (preferred kicking limb) and non-dominant limbs. Torque was calculated as the product of force (N) and shank length (m) relative to body mass (kg) and expressed as $Nm{\cdot}kg^{-1}$.

Season hamstring injuries

A hamstring injury was defined as acute pain in the posterior thigh that occurred during training or match-play, with accompanying symptoms of tenderness on palpation, pain on contraction and stretching, localized to the affected area (Van Dyk et al., 2017), which resulted in the player missing at least one match. This time-loss definition is consistent with previous AFL injury surveillance reports (Orchard & Seward, 2002; Orchard et al., 2013). Only hamstring injuries that occurred during the playing season were included in the study. Each team's senior physiotherapist was responsible for assessing and recording hamstring injuries during

Table 1. Participant characteristics measured at baseline for players that sustained a hamstring injury (injured group) and players that did not experience a hamstring injury (uninjured group) during the playing season.

	Uninjured group	Injured group	*P* value
N	111	14	
Age (years)	22 (3)	24 (4)	0.06
95% CI	21 to 23	22 to 27	
Height (cm)	189 (7)	188 (6)	0.59
95% CI	188 to 190	184 to 191	
Mass (kg)	87 (8)	87 (7)	0.96
95% CI	86 to 89	83 to 92	
AFL playing experience (years)	13 (5)	13 (7)	0.57
95% CI	12 to 13	9 to 17	
Professional AFL playing experience (years)	4 (3)	6 (4)	0.06
95% CI	3 to 4	4 to 9	

Table 2. Nordic strength and relative torque for the injured and uninjured limb classified from prospective and retrospective (≤12 months and during their career) hamstring injury data.

	Injured limb Mean (SD)	Uninjured limb Mean (SD)	Difference Mean (95% CI)	*P* value
Prospective hamstring injuries (n = 12)				
Force (N)	443 (87)	447 (72)	4 (−44 to 52)	0.85
Torque (Nm·kg^{-1})	2.3 (0.4)	2.3 (0.3)	0 (−0.2 to 0.3)	0.87
Retrospective hamstring injuries ≤ 12 months (n = 17)				
Force (N)	423 (86)	436 (73)	13 (−15 to 40)	0.35
Torque (Nm·kg^{-1})	2.1 (0.4)	2.1 (0.3)	0.1 (−0.1 to 0.2)	0.30
Retrospective hamstring injuries during their career (n = 35)				
Force (N)	422 (89)	429 (84)	7 (−9 to 22)	0.38
Torque (Nm·kg^{-1})	2.2 (0.4)	2.2 (0.4)	0 (0.0 to 0.1)	0.30

the playing season. As a minimum, each hamstring injury was assessed and reported clinically, with magnetic resonance imaging being undertaken in all cases except one. Post-injury, players were asked to report their mechanism of injury.

Statistical analysis

All statistical analyses were conducted using the Statistical Package for Social Sciences (version 25, IBM, USA). Data were checked for normality and assumptions were adhered to; therefore, parametric statistics were used. Independent samples *t*-test was used to assess the effect of group (season hamstring injury/no hamstring injury) on participant characteristics (age, height, mass, AFL playing experience, professional AFL playing experience, Nordic strength and torque). Paired samples *t*-tests were used to assess differences in Nordic strength and torque between subsequently injured and uninjured limbs, and between previously injured (less than 1-year and greater than 1-year) and uninjured limbs. Nordic strength and torque (both using the injured limb versus dominant limb for the uninjured group) were dichotomized using a receiver operator characteristic (ROC) curve to identify the value that maximized specificity and minimized sensitivity. Age was dichotomized using a cut-off value equal to or greater than 25 years, consistent with previous literature (Gabbe et al., 2006). Injury history was already dichotomized with responses of yes/no. The association of Nordic strength, torque, age and injury history with prospective hamstring injury during the AFL season was assessed using chi-square analysis and reported as Odds Ratios (ORs). Finally, an analysis of variance with Bonferroni post hoc was used to assess the effect of team on demographic, Nordic strength and hamstring injury information. Significance was accepted as $p < 0.05$. Data are presented as means (one SD) or mean difference (95% confidence interval, CI).

Results

Participant characteristics and prospective hamstring injury information

Fourteen players sustained a hamstring injury during the playing season. Players deemed fit to undertake the NHE by their respective medical departments (n = 112) participated in Nordic strength data collection, which resulted in 12 players with a hamstring injury and 100 players without a hamstring injury. Five hamstring injuries occurred in the dominant limb and seven hamstring injuries occurred in the non-dominant limb. One player sustained hamstring injuries to both limbs (on two separate occasions), another player sustained one recurrence, and on both occasions the limb of the initial injury was used as their injured limb (non-dominant limb for the player with a bilateral injury and dominant limb for the player with a recurrent injury). The biceps femoris long head muscle (BF_{LH}) was injured in isolation on 73% of occasions, with the remainder of occasions comprising BF_{LH} and biceps femoris short head, or semimembranosus with or without the semitendinosus. Running was the reported mechanism of injury on 64% of occasions, with multi-modal aspects of running, such as running and lunging, or running while bent over to pick up a ball and pushed in the back, and acceleration, forming the basis of these injuries. The remainder of occasions comprising jumping, stretching and kicking. The average number of games missed was 3 (2) with a range of 1–8.

There were no significant differences between subsequently injured and uninjured players in terms of height, mass or AFL playing experience ($p > 0.05$); however, age and professional AFL playing experience approached significance ($p = 0.06$). There were no significant differences in Nordic strength between the subsequently injured and the dominant (mean difference = 37 N, 95% CI = −19 to 94 N, p = 0.19) and non-dominant limbs (mean difference = 47 N, 95% CI = −11 to 104 N, p = 0.11) of the uninjured group. Similarly, there was no significant differences in torque between the subsequently injured and dominant (mean difference = 0.2 Nm·kg^{-1} 95% CI = −0.1 to 0.4 Nm·kg^{-1}, p = 0.20) and non-dominant limbs (mean difference = 0.2 Nm·kg^{-1} 95% CI = −0.1 to 0.5 Nm·kg^{-1}, p = 0.10) of the uninjured group.

Nordic strength and torque for injured and uninjured limbs

There were no significant differences in Nordic strength or torque between the injured and uninjured limbs when hamstring injuries were classified using prospective and retrospective data (*P* values all > 0.05; table 2).

Association of Nordic strength and torque, age and injury history with prospective hamstring injury

ROC curve analysis revealed a Nordic strength value of 415 N maximized specificity and minimized sensitivity that was associated with hamstring injury (area under curve = 0.595,

Table 3. OR for association of future hamstring injury by risk factor.

Risk factor	Number of unexposed players	Number (percentage) of unexposed players with hamstring injury	Number of exposed players	Number (percentage) of exposed players with hamstring injury	Odds Ratio (95% CI)	*P* value
Nordic strength (>415 N)	55	4 (7.3%)	57	8 (14.0%)	1.9 (0.6 to 6.0)	0.36
Torque (>2.1 Nm·kg^{-1})	46	3 (6.5%)	66	9 (13.6%)	2.1 (0.6 to 7.3)	0.35
Age (≥25 years)	93	7 (7.5%)	32	7 (21.9%)	2.9 (1.1 to 7.6)	<0.05[a]
Prior hamstring injury (≤1-year)	104	8 (7.7%)	21	6 (28.6%)	3.7 (1.4 to 9.6)	0.01[a]
Prior hamstring injury (>1-year)	83	5 (6.0%)	41	9 (22.0%)	3.6 (1.3 to 10.2)	0.01[a]
Prior groin injury (ever)	88	3 (3.4%)	34	10 (29.4%)	8.6 (3.5 to 29.4)	<0.01[a]
Prior calf injury (ever)	96	5 (5.2%)	25	6 (24.0%)	4.6 (1.5 to 13.9)	0.01[a]
Prior quadriceps injury (ever)	82	9 (11%)	39	3 (7.7%)	0.7 (0.2 to 2.4)	0.75
Prior knee injury (ever)	64	7 (10.9%)	56	5 (8.9%)	0.8 (0.3 to 2.4)	0.77
Prior ACL injury (ever)	89	11 (12.4%)	8	1 (12.5%)	N/A	

Exposed means in the nominally "high risk" category: Nordic strength > 415 N, Age ≥ 25 years, previous hamstring, groin, calf, quadriceps, knee, or ACL injury.
adenotes significant difference between the subsequently uninjured and subsequently injured players.

p = 0.28, Supplementary Figure 1), which was subsequently used to dichotomize Nordic strength (>415 N or <415 N) for the chi-square analysis. Based on this cut-off (415 N), no association was found between Nordic strength and increased hamstring injury risk (table 3). ROC curve analysis yielded a torque cut-off of 2.1 Nm·kg^{-1} (Supplementary Figure 1), which when used to dichotomize torque data (>2.1 Nm·kg^{-1} or <2.1 Nm·kg^{-1}), resulted in no association between torque and hamstring injury (table 3). However, age greater than 25 years (OR = 2.9, $p < 0.05$), a hamstring injury in the past 12 months (OR = 3.7, p = 0.01), or greater than 1-year ago (OR = 3.6, p = 0.01), or a prior groin (OR = 8.6, $p < 0.01$) or calf injury (OR = 4.6, p = 0.01) were significantly associated with the risk of future hamstring injury during the playing season (table 3). Of the 111 players who did not sustain a hamstring injury within the playing season, 52% (n = 57) reported a history of hamstring, groin or calf injury, whereas of the 14 players who sustained a hamstring injury during the playing season, 79% (n = 11) reported a history of hamstring, groin or calf injury. A history of quadriceps (OR = 0.7, p = 0.75) or knee (OR = 0.8, p = 0.77) injury was not associated with a future hamstring injury (table 3).

Between team differences

There were significant differences in Nordic strength between the individual teams ($p < 0.05$), whereas no significant differences were demonstrated for any demographic data (P values all > 0.05, Table 1).

Discussion

The aim of this study was to determine if previously identified risk factors for hamstring injury, such as low levels of Nordic strength, increasing age and prior lower limb injury, were associated with subsequent hamstring injury within an independent sample of elite AFL players. This study demonstrated that pre-season Nordic strength and torque were not associated with hamstring injury in the ensuing AFL playing season. In contrast, older players (≥25 years), and those with a history of hamstring, groin or calf injury, had significantly increased risk of sustaining a subsequent hamstring injury. These findings supported our hypotheses.

Nordic strength and torque

Nordic strength and corresponding torque values were not significantly associated with a future hamstring injury. Further, when participants were dichotomized by age, and according to retrospective and prospective injury, no significant differences in Nordic strength or corresponding torque values were detected. The lack of association between strength and hamstring injury in the present study are in agreement with Ruddy et al. (2018b), but contrast those of Opar et al. (2015) who reported a 4.3-fold greater risk of sustaining a hamstring injury in AFL players with Nordic strength values less than 279 N. It is notable that the Nordic strength values reported by Opar et al. (2015), with mean values between 284 and 330 N at the end of pre-season, were lower than the mean values reported by Ruddy et al. (2018b) with a mean (SD) of 341(80) N, and those observed in the present study with a mean (SD) of 409 (91) N. In fact, only seven players from the present study had Nordic strength below the 279 N threshold used to classify participants into low and high strength in the study by Opar et al. (2015), and none of the players within the current study who sustained a hamstring injury had Nordic strength values less than 279 N. The relative torque values for Nordic strength within our study were higher than those reported in professional soccer players (Timmins et al., 2016) and this is most likely due to the higher absolute Nordic strength values within the AFL players. The greater strength of players in the present study may reflect the training practices of the three

participating teams, which all incorporated Nordic strength training at least twice per week during the pre-season and once per week during the playing season, and are also consistent with a general trend in the AFL for increasing Nordic strength (Vald Performance, 2019). The Nordic strength values derived from the present study were also higher than mean values reported in studies of professional soccer (Timmins et al., 2016; Van Dyk et al., 2017), rugby union (Bourne et al., 2015), and Gaelic football players (Roe et al., 2020). These previous studies report conflicting evidence for an association between strength and hamstring injury. Taken together, these data suggest that the Nordic strength and torque of AFL players in the present study are higher compared with prior studies in AFL players (Opar et al., 2015; Ruddy et al., 2018b) and other sports (Roe et al., 2020; Timmins et al., 2016; Van Dyk et al., 2017) and evidence for an association between pre-season measures of Nordic strength, torque and future hamstring injury appears limited. This is further supported by the findings of a recent meta-analysis which reported that strength measured during the NHE was not associated with a future hamstring injury (Green et al., 2020).

Prior lower limb injury

Players who reported a prior hamstring injury (within the previous year or greater than 1-year), or a previous groin or calf injury, had approximately 4–8 times increased risk of sustaining a hamstring injury within the AFL playing season compared with those that had not sustained a prior injury. These findings are in general agreement with those of other AFL studies which reported that players with a history of hamstring injury within the previous year or greater than 1-year, were between 2.4 and 13 times more likely to suffer a recurrence than those with no history (Gabbe et al., 2006; Orchard, 2001; Orchard et al., 2020; Verrall et al., 2001). Similar findings have also been reported in soccer players with a hamstring injury within the previous season resulting in 1.6 times greater risk of hamstring injury (Engebretsen et al., 2010; Hägglund et al., 2013, 2006). Unlike Orchard et al. (2020) who reported a recent hamstring injury was more likely to be associated with a future hamstring injury than longer-term injuries, the risk of hamstring injury in the present study was similar for hamstring injuries in the past 12 months (OR = 3.7) compared with greater than 1-year ago (OR = 3.6). The higher OR of prior groin injury predicting subsequent hamstring injury may be due to the fact that AFL players may continue to play with low-grade/chronic groin pain, which can result in changes in motor unit recruitment strategy (Tucker & Hodges, 2010), whereas players typically do not continue to play with hamstring pain. Furthermore, the multi-directional and high-speed nature of AFL compared with other sports may result in greater physiological and mechanical stress on the adductor and the hamstring muscles, with the hamstring muscles more likely to succumb to injury during high-speed sagittal plane running; however, further work is needed to clarify this. A more granular record of the timing of prior injury is recommended in future studies to better address this question.

Evidence of greater metabolic muscle activity and decreased time to exhaustion within previously injured hamstring muscles compared with uninjured hamstring muscles is apparent during a leg-curl manoeuvre, even up to 6 (4) months post-injury (Schuermans et al., 2014), and this may explain, at least in part, the elevated risk for hamstring re-injury demonstrated shortly after return to play within the AFL (Orchard, 2001; Orchard et al., 2020) and soccer (Hägglund et al., 2013). It has been demonstrated that AFL players older than 24 years have a concomitant increase in hamstring and calf injury incidence, whereas no increase was demonstrated for quadriceps and knee injuries, and this has been attributed to degeneration within the respective lumbar spine nerve supply (Orchard et al., 2004). Several explanations have been postulated for the presence of a previous anatomically unrelated injury significantly increasing the risk of sustaining a hamstring injury. These include deficits in physical conditioning, proprioception or altered movement patterns as a direct result of the injury (Hägglund et al., 2006), or a lack of adequate rehabilitation and/or premature return to play (Hägglund et al., 2006) although there is limited research evidence to substantiate these arguments at present. Collectively, these findings indicate that rehabilitation of lower limb soft-tissue injury may be required long after return to sport has been achieved to reduce the occurrence of a future hamstring injury.

Age

Players aged 25 years or more demonstrated a significantly increased risk of future hamstring injury compared with younger players. In agreement with these findings, age greater than 23–24 years has previously been associated with sustaining a hamstring injury in AFL players (Gabbe et al., 2006; Orchard, 2001; Verrall et al., 2001), although no association has been reported in more recent studies investigating AFL players (Opar et al., 2015; Orchard et al., 2020). This discrepancy was due to the effect of age on future hamstring injury being confounded by injury history (Orchard et al., 2020). In contrast, another recent meta-analysis, which included a wide variety of sports, demonstrated increasing age as a significant risk factor in subsequent hamstring injury (Green et al., 2020). Increasing age may be a risk factor for injury because it correlates with exposure, as older players are exposed to a greater accumulation of physiological and mechanical stress over time and the likelihood of encountering an inciting event is increased.

This study had several limitations. Firstly, participants were recruited from three professional AFL teams, which may limit the generalizing of the findings since the association with hamstring injury may be influenced by between team differences in player profiles, e.g., age, prior injury, injury rates, hamstring strength and training practices. Indeed, we noted that there was some heterogeneity in the distribution of Nordic strength and hamstring injury rates between the three teams (see Table 1), but we did not run a within or between team or multivariate analysis due to low numbers, and the potential of interaction limiting our findings. Overcoming the multiple highly correlated variables with low study numbers may be achieved by using partial least squares correlation analysis (Weaving et al., 2019), which may be an area for future research. The benefit of this type of analysis is that many potential risk factors, such as the volume

of high-speed running (Ruddy et al., 2018a), may also be added to this model. Further, although Nordic strength data were collected by different personnel at each club, our standardized measurement approach and the high reliability of Nordic strength assessment (Opar et al., 2013) suggest that systematic error was minimized. Finally, the self-reported nature of the questionnaire is another limitation as this is dependent upon memory recall. Future prospective studies of hamstring injury risk in the AFL could consider the influence of factors that could change throughout the season, e.g., other injuries and training load, and the potential for injury risk factors to interact.

Conclusions

Nordic strength of professional AFL players appears to have increased over recent AFL seasons but was not found to be independently associated with hamstring injury. Consistent with prior literature, increasing age and prior hamstring, groin and calf injury remain strongly associated with future hamstring injury at the current time. Within the AFL, players with an elevated hamstring injury risk may be better identified from lower limb injury history data rather than by achieving maximum Nordic strength. Future efforts to decrease hamstring injury in the AFL should focus on better understanding the causal mechanisms underlying these well-established risk factors and their interactions.

Disclosure statement

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